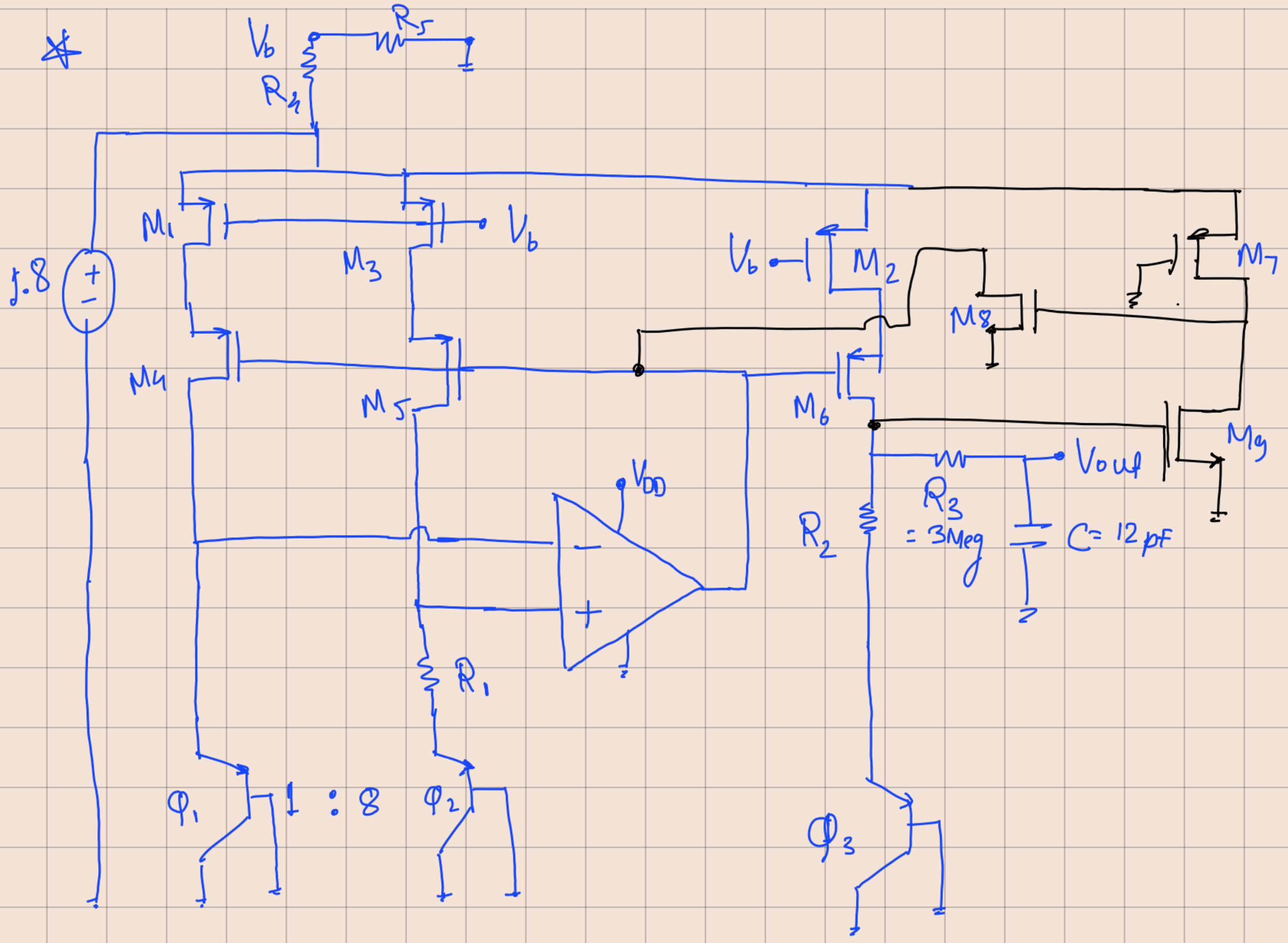
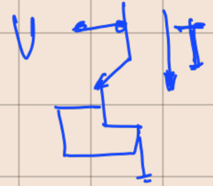
*



* Calculations

- Supply voltage = 1.8 V
- NMOS : $V_{TN} = 0.37V$, $\mu_n C_{ox} = 230\ \mu A/V^2$
- PMOS : $V_{Tp} = -0.39V$, $\mu_p C_{ox} = 100\ \mu A/V^2$
- $L_{min} = 0.18\ \mu m$, $W_{min} = 0.27\ \mu m$
- And BJT ratio $\{n\} = 8$

→ Generating CTAT



$$\frac{\partial V_{EB}}{\partial T} \approx -2mV/K$$

→ Generating PTAT

$$\Delta V_{EB} = V_{EB_2} - V_{EB_1} = V_T \ln(n)$$

$\therefore \; I_{PTAT} = \dfrac{V_T \ln(n)}{R_1}$

→ This current is mirrored into the third branch & flows through R_2 & Q_3

$$\therefore \; V_{ref} = V_{EB_3} + \left(\frac{R_2}{R_1}\right) V_T \ln n$$

$$\frac{\partial V_{ref}}{\partial T} = \frac{\partial V_{EB_3}}{\partial T} + \left(\frac{R_2}{R_1}\right) \ln(n) \frac{\partial V_T}{\partial T} = 0$$

→ $\dfrac{\partial V_{EB}}{\partial T} = -2\,mV/K$

→ $\dfrac{\partial V_T}{\partial T} = \dfrac{k}{q} = 0.0862\ mV/K$

→ $n = 8 \Rightarrow \ln(8) = 2.079$

$$\therefore \; R_2 / R_1 \approx 11.17$$

⟹ Design Choice ⇒ $I_{PTAT} = 10\,\mu A$

V_T @ $300K \approx 26\,mV$

$\therefore \; \Delta V_{EB} = 26mV \cdot \ln(8) = 53.74\,mV$

$$R_1 = \frac{\Delta V_{EB}}{I_{PTAT}} = 5.374\ k\Omega$$

$$\therefore \; R_2 = 11.17 \times R_1 = 60.03\,k\Omega$$

And $V_{ref_{theoretical}} = V_{EB_3} + I R_2$

$= 0.65 + 10 \times 10^{-6} \times 60.03 \times 10^3$

$$\boxed{V_{ref_{theoretical}} = 1.25\,V}$$

✩ Biasing of transistors

→ To provide maximum voltage headroom for cascodes & the BJTs below, M_1, M_2, M_3 must act as small voltage-dropping resistors, operating deep in the linear (triode) region.

→ $V_G = \frac{V_{DD}}{2} = 0.9V$

→ This requires $R_4 = R_5$. To keep static power waste low, we select $R_4 = R_5 = 500\,k\Omega$

✩ Transistor sizing

→ $V_{SG} = 1.8 - 0.9 = 0.9V$

$V_{SD} = 0.1V$ {for deep triode}

$\left\{ \begin{array}{c} V_{SD} < (V_{SG} - |V_{TP}|) \\ 0.2 < 0.51 \end{array} \right\}$

∴ linear

$I_D \approx \mu_p C_{ox} \left(\frac{W}{L}\right) \left[(V_{SG} - |V_{TP}|\, V_{SD}) \right]$

$10\,\mu A = 100\,\mu A/V^2 \left(\frac{W}{L}\right) \left[(0.9 - 0.39) \cdot 0.1 \right)$

$\Rightarrow \boxed{W/L = \frac{10}{5.1} \approx 1.96}$

∴ $W = 2\,\mu m$ & $L = 1\,\mu m$ for M_1, M_2, M_3

→ for cascodes

— source node is at $1.8 - 0.2 = 1.6$

$$I_D = \frac{1}{2}\mu_p C_{ox} \frac{W}{L} \times V_{ov}^2 \quad \{V_{ov} = 0.2V\}$$

$$10\mu A = \frac{1}{2} \times (100\mu A/V^2)\left(\frac{W}{L}\right)(0.2V)^2$$

$$\Rightarrow \boxed{W/L = 5}$$

$\therefore W = 5\mu m$, $L = 1\mu m$ for M_4, M_5, M_6

☆ <u>Output filter</u>

$$f_c = \frac{1}{2\pi R_3 C_1} = \frac{1}{2 \times \pi \times 3 \times 10^6 \times 12 \times 10^{-12}}$$

$$= \frac{10^6}{72 \times \pi} \approx 4.42 \text{ kHz}$$

$\therefore$ High freq. noise rejected